Exercise list — Topic 4 (A and B): Moral hazard, credit rationing, asset substitution, debt overhang, renegotiation

One list for Prof. Ricca's Topic 4A (`Aula/Topic4_A.pdf`: the Holmström–Tirole fixed-investment model, pledgeable income, credit rationing, debt versus equity) and Topic 4B (`Aula/Topic4_B.pdf`: asset substitution, debt overhang, renegotiation and bankruptcy design), mapped in lectures-index. 15 real exercises ordered easy → really hard, copied verbatim from the course books for personal study; pages are printed, with the PDF page in brackets. Every block is taken unchanged from lists R4 and R7, where it already passed `tools/check_lists.py`; the last column says where. No book exercise covers Indarte's default-as-insurance model or the Karlan–Zinman experiment. Links: exercise index · rules · R4 · R7.

Lecture block	Exercises
4A.2–4A.4 Fixed-investment model, rationing	5, 8, 9
4A.5 Investment and internal funds	11
4A.7 Debt dominates equity	10
4A.4 Collateral reading	13
4B.1–4B.3 Asset substitution and remedies	2, 4, 6, 14
4B.4 Debt overhang	1, 3, 7
4B.5–4B.6 Renegotiation and bankruptcy design	12, 15

References

#	Tier	Book	Ch.	Exercise	Printed p. (PDF)	Answer in book	Trains	From
1	Easy	Grinblatt & Titman 2e	Ch. 16	16.14	593 (606)	no	Senior DIP financing as a cure for debt overhang that helps existing creditors	R7 ex. 2
2	Easy	Grinblatt & Titman 2e	Ch. 16	16.1	591 (604)	no	Asset substitution: equity prefers the negative-NPV risky project when debt is due	R7 ex. 4
3	Easy	Berk–DeMarzo–Harford 2e	Ch. 16	16.25	494–495 (525)	no	Debt overhang: positive-NPV project that equity will not fund	R4 ex. 4
4	Easy	Grinblatt & Titman 2e	Ch. 16	16.11	593 (606)	no	Debt maturity as a cure for debt holder–equity holder incentive problems	R4 ex. 2
5	Easy	Tirole	Ch. 3	3.1	144 (157)	yes, p. 571	Fixed-investment model, breakeven condition, random financing below $\bar{A}$	R4 ex. 5
6	Exam	Berk–DeMarzo–Harford 2e	Ch. 16	16.26*	495 (526)	no	Leverage-driven strategy choice and the expected agency cost of debt	R7 ex. 7
7	Exam	Grinblatt & Titman 2e	Ch. 16	16.12 + 16.13	593 (606)	no	Debt overhang: positive-NPV project that equity cannot fund over a senior bond; riskier variant	R7 ex. 8
8	Exam	Tirole	Ch. 3	3.15	150 (163)	yes, p. 578	Pledgeable income and riskiness: which variant is less rationed	R4 ex. 6
9	Exam	Tirole	Ch. 3	3.2	144 (157)	yes, p. 571	Minimum equity under entrepreneurial risk aversion, with/without limited liability	R4 ex. 9
10	Exam	Tirole	Ch. 3	3.18	151 (164)	yes, p. 579	Maximal incentives principle: optimality of the debt contract	R4 ex. 10

#	Tier	Book	Ch.	Exercise	Printed p. (PDF)	Answer in book	Trains	From
11	Exam	Tirole	Ch. 3	3.11	148 (161)	yes, p. 575	Shadow value of retained earnings; cash flow and future financing	R4 ex. 11
12	Really hard	Tirole	Ch. 3	3.6	146 (159)	yes, p. 573	Renegotiation before effort: when lenders forgive debt to restore (IC_b) in the variable-investment model	R7 ex. 14
13	Really hard	Tirole	Ch. 4	4.17	195 (208)	yes, p. 591	Comparative statics of investment size and failure-contingent collateral with concave $R(I)$ and $\phi(C)$	R7 ex. 16
14	Really hard	Tirole	Ch. 7	7.3	323–324 (336)	yes, p. 602	Pledgeable liquidation value, asset substitution via hidden maintenance choice, minimum A that neutralizes it	R7 ex. 17
15	Really hard	Tirole	Ch. 4	4.9	190–191 (203)	yes, p. 586	Optimal continuation cutoff r^* as short-term debt; why a borrower-friendly court hurts the borrower	R7 ex. 18

Tier 1 — Easy

1. DIP financing and existing creditors

Source: Grinblatt, Ch. 16, Exercise 16.14, p. 593 (PDF 606) · Answer in book: no

With debtor in possession (DIP) financing, bankrupt firms are able to obtain additional amounts of debt that is senior to the firm's existing debt. Explain how the firm's existing debt holders can benefit from this.

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2. Asset substitution: project A or B

Source: Grinblatt, Ch. 16, Exercise 16.1, p. 591 (PDF 604) · Answer in book: no

A firm has $100 million in cash on hand and a debt obligation of $100 million due in the next period. With this cash, it can take on one of two projects—A or B—which cost $100 million each. Assume that the firm cannot raise any additional outside funds. If the economy is favorable, project A will pay $120 million and project B will pay $101 million. If the economy is unfavorable, project A will pay $60 million and project B will pay $101 million. Assume that investors are risk neutral, there are no taxes or direct costs of bankruptcy, the riskless interest rate is zero, and the probability of each state is .5.

a. What is the NPV of each project?

b. Which project will equity holders want the managers to take? Why?

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3. Debt overhang on vacant land

Source: Berk, Ch. 16, Problem 16.25, pp. 494–495 (PDF 525) · Answer in book: no

Consider a firm whose only asset is a plot of vacant land, and whose only liability is debt of $15 million due in one year. If left vacant, the land will be worth $10 million in one year. Alternatively, the firm can develop the land at an upfront cost of $20 million. The developed land will be worth $35 million in one year. Suppose the risk-free interest rate is 10%, assume all cash flows are risk-free, and assume there are no taxes.

a. If the firm chooses not to develop the land, what is the value of the firm's equity today? What is the value of the debt today?

b. What is the NPV of developing the land?

c. Suppose the firm raises $20 million from equity holders to develop the land. If the firm develops the land, what is the value of the firm's equity today? What is the value of the firm's debt today?

d. Given your answer to part (c), would equity holders be willing to provide the $20 million needed to develop the land?

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4. Short-term debt and incentive problems

Source: Grinblatt, Ch. 16, Exercise 16.11, p. 593 (PDF 606) · Answer in book: no

Why are debt holder–equity holder incentive problems less severe for firms that borrow short term rather than long term?

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5. Random financing

Source: Tirole, Ch. 3, Exercise 3.1, p. 144 (PDF 157) · Answer in book: yes, p. 571

Exercise 3.1 (random financing). Consider the fixed-investment model of Section 3.2. We know that if $A \geqslant \bar{A}$, where

$$I - \bar{A} = p_H \left(R - \frac{B}{\Delta p} \right),$$

it is both optimal and feasible for the borrower to sign a contract in which the project is undertaken for certain. We also noted that for $A < \bar{A}$, the borrower cannot convince investors to undertake the project with probability 1. With $A > 0$, the entrepreneur benefits from signing a "random financing contract," though.

(i) Consider a contract in which the borrower invests $\hat{A} \in [0, A]$ of her own money, the project is financed with probability x, and the borrower receives R_b in the case of success and 0 otherwise. Write the investors' breakeven condition.

(ii) Show that (provided the NPV, $p_H R - I$, is positive) it is optimal for the borrower to invest

$$\hat{A} = A.$$

How does the probability that the project is undertaken vary with A?

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. […] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. […]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") […].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. […]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure […]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[…] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[…] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

[…] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. […]

Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. […] There is *credit rationing.* […]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I-A}{p_H} \geqslant R - \frac{I-\bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

[…] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H(R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

So, the borrower's utility jumps up at $A = \bar{A}$. […]

From Tirole §3.2.1, footnote 9, p. 116 (PDF 129):

"Random financing" contracts, in which the borrower brings equity in exchange for a probability between 0 and 1 of being financed may in some cases be optimal when the investment size is fixed (as it is here) or more generally in the presence of indivisibilities or increasing returns to scale (see Exercise 3.1). For simplicity, we focus on deterministic contracts.

Tier 2 — Exam level

6. Zymase: leverage and strategy choice

Source: Berk, Ch. 16, Problem *26, p. 495 (PDF 526) · Answer in book: no

*26. Zymase is a biotechnology start-up firm. Researchers at Zymase must choose one of three different research strategies. The payoffs (after taxes) and their likelihood for each strategy are shown below. The risk of each project is diversifiable.

Strategy	Probability	Payoff ($ million)
A	100%	75
B	50%	140
	50%	0
C	10%	300
	90%	40

a. Which project has the highest expected payoff?

b. Suppose Zymase has debt of $40 million due at the time of the project's payoff. Which strategy has the highest expected payoff for equity holders?

c. Suppose Zymase has debt of $110 million due at the time of the strategy's payoff. Which strategy has the highest expected payoff for equity holders?

d. If management chooses the strategy that maximizes the payoff to equity holders, what is the expected agency cost to the firm from having $40 million in debt due? What is the expected agency cost to the firm from having $110 million in debt due?

7. Ajax Manufacturing: debt overhang

Source: Grinblatt–Titman, Ch. 16, Exercises 16.12 and 16.13, p. 593 (PDF 606) · Answer in book: no

16.12. Consider the case of Ajax Manufacturing which just completed an R&D project on widgets that required a \$70 million bond obligation. The R&D effort resulted in an investment opportunity that will cost \$75 million and generate cash flows of \$85 million in the event of a recession (*prob.* = 20%) and \$150 million if economic conditions are favorable (*prob.* = 80%). What is the *NPV* of the project assuming no taxes, no direct bankruptcy costs, risk neutrality, and a risk-free interest rate of zero? Can the firm fund the project if the original debt is a senior obligation that doesn't allow the firm to issue additional debt?

16.13. Assume now that if Ajax Manufacturing (see exercise 16.12) uses a more capital-intensive manufacturing process, it can produce a greater number of widgets at a lower variable cost. Given the greater fixed costs, the cash flows are only \$5 million in an unfavorable economy with the capital-intensive process but are \$170 million in a favorable economy. Hence, equity holders would receive \$100 million in the good state of the economy (\$170 million − \$70 million) and zero in a recession because \$5 million is less than the \$70 million debt obligation. Can the firm issue equity to fund the project?

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8. Project riskiness and credit rationing

Source: Tirole, Ch. 3, Exercise 3.15, p. 150 (PDF 163) · Answer in book: yes, p. 578

Exercise 3.15 (project riskiness and credit rationing). Consider the basic, fixed-investment model (the investment is I, the entrepreneur borrows $I - A$; the probability of success is p_H (no private benefit) or $p_L = p_H - \Delta p$ (private benefit B), success (failure) yields verifiable profit R (respectively 0)). There are two variants, "A" and "B," of the projects, which differ only with respect to "riskiness":

$$p_H^A R^A = p_H^B R^B, \quad \text{but } p_H^A > p_H^B;$$

so project B is "riskier." The investment cost is the same for both variants and, furthermore,

$$p_H^A - p_L^A = p_H^B - p_L^B.$$

Which variant is less prone to credit rationing?

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. […] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. […]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") […].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. […]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will

imply that both sides receive 0 in the case of failure […]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[…] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[…] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

[…] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. […]

Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. […] There is *credit rationing.* […]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I - A}{p_H} \geqslant R - \frac{I - \bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

[…] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H(R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

So, the borrower's utility jumps up at $A = \bar{A}$. […]

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9. Entrepreneurial risk aversion

Source: Tirole, Ch. 3, Exercise 3.2, p. 144 (PDF 157) · Answer in book: yes, p. 571

Exercise 3.2 (impact of entrepreneurial risk aversion). Consider the fixed-investment model developed in this chapter: an entrepreneur has cash amount A and wants to invest $I > A$ into a project. The project yields $R > 0$ with probability p and 0 with probability $1 - p$. The probability of success is p_H if the entrepreneur works and $p_L = p_H - \Delta p$ ($\Delta p > 0$) if she shirks. The entrepreneur obtains private benefit B if she shirks and 0 otherwise. Assume that

$$I > p_H \left(R - \frac{B}{\Delta p} \right).$$

(Suppose that $p_L R + B < I$; so the project is not financed if the entrepreneur shirks.)

(i) In contrast with the risk-neutrality assumption of this chapter, assume that the entrepreneur has utility for consumption c:

$$u(c) = \begin{cases} c & \text{if } c \geqslant c_0, \\ -\infty & \text{otherwise.} \end{cases}$$

(Assume that $A \geqslant c_0$ to ensure that the entrepreneur is not in the "$-\infty$ range" in the absence of financing.)

Compute the minimum equity level $\bar{A}$ for which the project is financed by risk-neutral investors when the market rate of interest is 0. Discuss the difference between $p_H = 1$ and $p_H < 1$.

(ii) Generalize the analysis to risk aversion. Let $u(c)$ denote the entrepreneur's utility from consumption with $u' > 0$, $u'' < 0$. Conduct the analysis assuming either limited liability or the absence of limited liability.

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. […] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. […]

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Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. […]

Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. […] There is *credit rationing.* […]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I-A}{p_H} \geqslant R - \frac{I-\bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

[…] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H(R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

So, the borrower's utility jumps up at $A = \bar{A}$. […]

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10. Maximal incentives principle in the fixed-investment model

Source: Tirole, Ch. 3, Exercise 3.18, p. 151 (PDF 164) · Answer in book: yes, p. 579

Exercise 3.18 (maximal incentives principle in the fixed-investment model). Pursue the analysis of Section 3.4.3, but for the fixed-investment model of Section 3.2: the investment cost I is given and the income is either R^S or R^F (instead of R or 0), where $R^S > R^F > 0$. We assume that

$$R^F < I - A,$$

so the project cannot be straightforwardly financed by bringing in net worth A and pledging the lower income R^F to lenders. Let

$$R \equiv R^S - R^F$$

denote the increase in income from the low to the high level. Show that the debt contract is optimal, but unlike in the variable-investment case it may not be uniquely optimal.

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. […] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. […]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") […].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. […]

Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. […] There is *credit rationing.* […]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I-A}{p_H} \geqslant R - \frac{I-\bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

[…] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H(R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

So, the borrower's utility jumps up at $A = \bar{A}$. […]

From Tirole §3.4.3 "Collateral Values: Outside Debt and the Maximal Incentives Principle", pp. 128–129 (PDF 141–142):

We now return to the indeterminacy of the financial structure (debt or equity) discussed earlier. It turns out that this indeterminacy was an artefact of the absence of profit in the case of failure.

Thus, assume that, for investment size I, the profit is $R^S I$ in the case of success and $R^F I$ in the case of failure, where R^F is now positive. $R^F I$ can be thought of as the salvage value of assets and

$$RI \equiv (R^S - R^F)I$$

as the increase in profit brought about by success. […]

The model is otherwise the same as in the rest of Section 3.4: the private benefit (BI in the case of misbehavior, 0 otherwise) is also proportional to investment.

The generalization of the condition that the NPV per unit of investment is positive while the pledgeable income per unit of investment is negative ($p_H R > 1 > p_H(R - B/\Delta p)$) is

$$p_H R + R^F > 1 > p_H \left(R - \frac{B}{\Delta p} \right) + R^F.$$

A contract specifies an investment level I and a sharing rule, or equivalently a reward for the entrepreneur for each performance level: $\{R_b^S, R_b^F\}$, with $R_b^S, R_b^F \geqslant 0$ due to limited liability.

The optimal contract maximizes the entrepreneur's expected compensation,

$$U_b = \max_{\{R_b^S, R_b^F, I\}} \{p_H R_b^S + (1 - p_H) R_b^F - A\},$$

subject to two constraints (that will turn out to be binding at the optimum): the entrepreneur's incentive constraint,

$$(\Delta p)(R_b^S - R_b^F) \geqslant BI,$$

and the investors' breakeven constraint,

$$p_H(R^S I - R_b^S) + (1 - p_H)(R^F I - R_b^F) \geqslant I - A.$$

To show that the investors' breakeven constraint is binding, note that, if it were not, then the entrepreneur could increase R_b^S and R_b^F by an equal and small amount without affecting the incentive compatibility constraint. This uniform increase in compensation would raise the entrepreneur's payoff. As is now familiar, we conclude that the investors receive no surplus, and so (by substituting the breakeven constraint into the objective function) the entrepreneur's utility is equal to the NPV:

$$U_b = (p_H R + R^F - 1)I.$$

Because the NPV per unit of investment is positive, the entrepreneur therefore chooses the highest possible investment.

Next, note that the incentive constraint is binding (otherwise, the optimal investment would be infinite, which would violate the two constraints combined).

Lastly, suppose that $R_b^F > 0$ at the optimum. And consider a small increase $\delta R_b^S > 0$ in managerial compensation in the case of success together with a small decrease $\delta R_b^F < 0$ in the case of failure that keeps the investors' profitability constant:

$$p_H \delta R_b^S + (1 - p_H)\delta R_b^F = 0.$$

This small change (which is feasible only if $R_b^F > 0$) also keeps the objective function constant. But the incentive constraint is now slack, a contradiction. We thus conclude that at the optimum

$$R_b^F = 0.$$

Hence, an all-equity firm cannot be optimal: in the absence of debt, the entrepreneur would receive $R^F I$ times her share of stocks in the firm, and therefore would be rewarded even in the case of failure. By contrast, investors' holding debt $D \geqslant R^F I$ is an optimal financial structure. Using the fact that the two constraints are binding, the borrowing capacity is given by

$$R^F I + p_H \left(R - \frac{B}{\Delta p} \right) I = I - A$$

or

$$I = \frac{A}{1 - [p_H(R - B/\Delta p) + R^F]}. \qquad (3.16)$$

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11. Retained-earnings benefit

Source: Tirole, Ch. 3, Exercise 3.11, p. 148 (PDF 161) · Answer in book: yes, p. 575

Exercise 3.11 (retained-earnings benefit). An entrepreneur has at date 1 a project of fixed size with characteristics $\{I^1, R^1, p_H^1, p_L^1, B^1\}$ (see Section 3.2). This entrepreneur will at date 2 have a different fixed size project with characteristics $\{I^2, R^2, p_H^2, p_L^2, B^2\}$, which will then require new financing. So, we are considering only a short-term loan for the first project. Retained earnings from the first project can, however, be used to defray part of the investment cost of the second project. Assume that all the characteristics of the second project are known at date 1 except B^2, which is distributed on $[\underline{B}^2, \bar{B}^2]$ according to the cumulative distribution $F(B^2)$. Assume for simplicity that $\underline{B}^2 > \Delta p^2(p_H^2 R^2 - I^2)/p_H^2$. The characteristics of the second project become common knowledge at the beginning of date 2.

(i) Compute the shadow value of retained earnings. (Hint: what is the entrepreneur's gross utility in period 2?)

(ii) Show that it is possible that the first project is funded even though it would not be funded if the second project did not exist and even though the entrepreneur cannot pledge at date 1 income resulting from the second project.

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\mathrm{IR}_l)$$

[…] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. […]

Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. […] There is *credit rationing.* […]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I - A}{p_H} \geqslant R - \frac{I - \bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

[…] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H (R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

So, the borrower's utility jumps up at $A = \bar{A}$. […]

From Tirole §3.4.2, p. 128 (PDF 141) — definition of the shadow value of equity:

Another important concept (which will be used, for example, in computing the value of retained earnings in a dynamic context) is the *shadow value v of equity (here cash).* The entrepreneur derives gross utility $v > 1$ from one more unit of equity. Letting $U_b^g \equiv A + U_b$ denote the borrower's gross utility, […]

Tier 3 — Really hard

12. Renegotiation and debt forgiveness

Source: Tirole, Ch. 3, Exercise 3.6, p. 146 (PDF 159) · Answer in book: yes, p. 573

Exercise 3.6 (renegotiation and debt forgiveness). When computing the multiplier k (given by equation (3.12)), we have assumed that it is optimal to specify a stake for the borrower large enough that the incentive constraint (IC_b) is satisfied. Because condition (3.8) implies that the project has negative NPV in the case of misbehavior, such a specification is clearly optimal when the contract cannot be renegotiated. The purpose of this exercise is to check in a rather mechanical way that the borrower cannot gain by offering a loan agreement in which (IC_b) is not satisfied, and which is potentially renegotiated before the borrower chooses her effort. While there is a more direct way to prove this result, some insights are gleaned from this pedestrian approach. Indeed, the exercise provides conditions under which the lender is willing to forgive debt in order to boost incentives (the analysis will bear some resemblance to that of liquidity shocks in Chapter 5, except that the lender's concession takes the form of debt forgiveness rather than cash infusion).[60]

(i) Consider a loan agreement specifying investment I and stake $R_b < BI/\Delta p$ for the borrower. Suppose that the loan agreement can be renegotiated after it is signed and the investment is sunk and before the borrower chooses her effort. Renegotiation takes place if and only if it is mutually advantageous. Show that the loan agreement is renegotiated if and only if

$$(\Delta p)RI - \frac{p_H BI}{\Delta p} + p_L R_b \geqslant 0.$$

(ii) Interpret the previous condition. In particular, show that it can be obtained directly from the general theory. Hint: consider a fictitious, "fixed-investment" project with income $(\Delta p)RI$, investment 0, and cash on hand $p_L R_b$.

(iii) Assume for instance that the entrepreneur makes a take-it-or-leave-it offer in the renegotiation (that is, the entrepreneur has the bargaining power). Compute the borrowing capacity when $R_b < BI/\Delta p$ and the loan agreement is renegotiated.

(iv) Use a direct, rational expectations argument to point out in a different way that there is no loss of generality in assuming $R_b \geqslant BI/\Delta p$ (and therefore no renegotiation).

[60] The phenomenon of debt renegotiation has been analyzed in a number of settings: see, for example, Bulow and Rogoff (1989a,b), Eaton and Gersovitz (1981), Fernandez and Rosenthal (1990), Gale and Hellwig (1989), Gromb (1994), Hart and Moore (1989, 1995), and Snyder (1994).

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", p. 115 (PDF 128) (assumptions kept in the variable-investment model):*

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. […]

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130) (the "general theory" of the fixed-investment model used in part (ii); here I is the fixed investment, A the borrower's cash, R the income in the case of success and B the private benefit):

[…] Because she has stake R_b in the firm's income (she receives R_b in the case of success and 0 in the case of failure), the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p)R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

where "IR_l" stands for the lenders' individual rationality constraint (which we will also often call the "breakeven constraint" or the "participation constraint"). Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

From Tirole §3.4.1 "The Continuous-Investment Model", p. 127 (PDF 140):

The continuous-investment model of this section is the polar opposite of the fixed-investment one. The fixed-investment model depicts a situation in which returns are sharply decreasing beyond some investment level. In contrast, we now assume that there are constant returns to scale in the investment technology. An investment $I \in [0, \infty)$ yields income RI, proportional to I, in the case of success, and 0 in the case of failure. The borrower's private benefit from misbehaving is also taken to be proportional to investment. As before, the borrower has a choice between behaving, in which case she derives no private benefit and the probability of success is p_H, and misbehaving, that is, enjoying private benefit BI, and reducing the probability of success to $p_L = p_H - \Delta p < p_H$. [...]

The borrower initially has cash A, and must therefore borrow $I - A$ to finance a project of size I. A loan agreement specifies that the lenders (who as before are assumed to make no profit) and the borrower receive 0 each in the case of failure, and R_l and R_b, respectively, in the case of success, where $R_l + R_b = RI$.

As in Section 3.2, we assume that investment has positive NPV (net present value), here per unit of investment, if the borrower behaves,

$$p_H R > 1, \qquad (3.7)$$

but negative NPV otherwise,

$$1 > p_L R + B, \qquad (3.8)$$

so that unless one can control the agency problem the investment cannot be funded. We also make an assumption that guarantees that the equilibrium investment is finite:

$$p_H R < 1 + \frac{p_H B}{\Delta p}. \qquad (3.9)$$

Like inequality (3.5) in Section 3.2, inequality (3.9) has a simple interpretation: the expected net revenue per unit of investment, $p_H R - 1$, is lower than the per-unit agency cost, $p_H B / \Delta p$.

Finally, we keep assuming that the capital market is competitive. [...]

From Tirole §3.4.2 "The Lenders' Credit Analysis", pp. 127–128 (PDF 140–141):

Following the steps of Section 3.2, the incentive compatibility and the breakeven conditions are

$$(\Delta p) R_b \geqslant BI \qquad (\text{IC}_b)$$

and

$$p_H (RI - R_b) \geqslant I - A. \qquad (\text{IR}_l)$$

In equilibrium, competitive lenders make no profit on the contract that is most advantageous for the borrower; the borrower's net utility is therefore equal to the social surplus brought about by the investment:

$$U_b = (p_H R - 1)I. \qquad (3.10)$$

From (3.10) it is optimal for the borrower to invest as much as possible. The upper bound on investment and in turn her borrowing capacity ("outside financing capacity" or "debt capacity") are determined by constraints (IC_b) and (IR_l). Substituting (IC_b) into (IR_l), we obtain

$$I \leqslant kA, \qquad (3.11)$$

where

$$k = \frac{1}{1 - p_H(R - B/\Delta p)} > 1. \qquad (3.12)$$

The denominator of k is positive from (3.9). Furthermore, conditions (3.7) and (3.8) imply that $(\Delta p)R > B$, and therefore that the denominator of k is smaller than 1. This is important: the fact that $k > 1$ shows that the borrower can lever her wealth, k being the multiplier.

[…] Conditions (3.7) and (3.10) furthermore imply that it is optimal for the borrower to invest k times her cash A, that is, to borrow $d = (k - 1)$ times her level of cash, where

$$d = \frac{p_H(R - B/\Delta p)}{1 - p_H(R - B/\Delta p)}. \qquad (3.13)$$

The maximum loan, dA, is called "borrowing capacity." […]

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13. Loan size and collateral requirements

Source: Tirole, Ch. 4, Exercise 4.17, p. 195 (PDF 208) · Answer in book: yes, p. 591

Exercise 4.17 (loan size and collateral requirements). An entrepreneur with limited wealth A finances a variable-investment project. A project of size $I \in \mathbb{R}$ if successful yields $R(I)$, where $R(0) = 0$, $R' > 0$, $R'' < 0$, $R'(0) = \infty$, $R'(\infty) = 0$. The probability of success is p_H if the entrepreneur behaves (she then receives no private benefit) and $p_L = p_H - \Delta p$ if she misbehaves (she then receives private benefit BI).

The entrepreneur can pledge an arbitrary amount of collateral with cost $C \geqslant 0$ to the entrepreneur and value $\phi(C)$ for the investors with $\phi(0) = 0$, $\phi' > 0$, $\phi'' < 0$, $\phi'(0) = 1$, $\phi'(\infty) = 0$.

The entrepreneur is risk neutral and protected by limited liability and the investors are competitive, risk neutral, and demand a rate of return equal to 0.

Assume that the first-best policy does not yield enough pledgeable income. (This first-best policy is $C^* = 0$ and I^* given by $p_H R'(I^*) = 1$. Thus, the assumption is $p_H[R(I^*) - BI^*/\Delta p] < I^* - A$.)

Assume that the entrepreneur pledges collateral only in the case of failure (on this, see Section 4.3.5), and that the investors' breakeven constraint is binding. Show that as A decreases or the agency cost (as measured by B or, keeping p_H constant, $p_H/\Delta p$) increases, the optimal investment size decreases and the optimal collateral increases.

Context needed to solve it *From Tirole §3.4.1 "The Continuous-Investment Model", p. 127 (PDF 140):*

The continuous-investment model of this section is the polar opposite of the fixed-investment one. […] we now assume that there are constant returns to scale in the investment technology. An investment $I \in [0, \infty)$ yields income RI, proportional to I, in the case of success, and 0 in the case of failure. The borrower's private benefit from misbehaving is also taken to be proportional to investment. As before, the borrower has a choice between behaving, in which case she derives no private benefit and the probability of success is p_H, and misbehaving, that is, enjoying private benefit BI, and reducing the probability of success to $p_L = p_H - \Delta p < p_H$. […]

The borrower initially has cash A, and must therefore borrow $I - A$ to finance a project of size I. A loan agreement specifies that the lenders (who as before are assumed to make no profit) and the borrower receive 0 each in the case of failure, and R_l and R_b, respectively, in the case of success, where $R_l + R_b = RI$.

As in Section 3.2, we assume that investment has positive NPV (net present value), here per unit of investment, if the borrower behaves,

$$p_H R > 1, \qquad (3.7)$$

but negative NPV otherwise,

$$1 > p_L R + B, \qquad (3.8)$$

so that unless one can control the agency problem the investment cannot be funded. We also make an assumption that guarantees that the equilibrium investment is finite:

$$p_H R < 1 + \frac{p_H B}{\Delta p}. \qquad (3.9)$$

[…] Finally, we keep assuming that the capital market is competitive. […]

From Tirole §3.4.2 "The Lenders' Credit Analysis", pp. 127–128 (PDF 140–141):

Following the steps of Section 3.2, the incentive compatibility and the breakeven conditions are

$$(\Delta p) R_b \geqslant BI \qquad (\mathrm{IC}_b)$$

and

$$p_H (RI - R_b) \geqslant I - A. \qquad (\mathrm{IR}_l)$$

In equilibrium, competitive lenders make no profit on the contract that is most advantageous for the borrower; the borrower's net utility is therefore equal to the social surplus brought about by the investment:

$$U_b = (p_H R - 1) I. \qquad (3.10)$$

From (3.10) it is optimal for the borrower to invest as much as possible. The upper bound on investment and in turn her borrowing capacity ("outside financing capacity" or "debt capacity") are determined by constraints (IC_b) and (IR_l). Substituting (IC_b) into (IR_l), we obtain

$$I \leqslant kA, \qquad (3.11)$$

where

$$k = \frac{1}{1 - p_H (R - B/\Delta p)} > 1. \qquad (3.12)$$

[…] the fact that $k > 1$ shows that the borrower can lever her wealth, k being the multiplier. […]

From Tirole §4.3.5 "Pledging Existing Wealth", p. 170 (PDF 183):

[…] Suppose that the entrepreneur can pledge an arbitrary amount C,

$$0 \leqslant C \leqslant C^{\max},$$

conditional on failing (we will later check that conditional collateral dominates unconditional collateral). Investors value collateral C at βC, where $\beta < 1$, when they seize it.

The borrower's net utility, as usual, is equal to the NPV. The NPV is equal to its value in the absence of collateral, $p_H R - I$, minus the deadweight loss associated with collateral pledging. This deadweight loss is equal to $(1 - \beta)C$ times the probability, $1 - p_H$, that the firm fails. And so

$$U_b = p_H R - I - (1 - p_H)(1 - \beta)C.$$

The NPV is maximized when $C = 0$; the borrower will therefore not pledge collateral unless she needs to. Thus, if A denotes the borrower's cash on hand and $A \geqslant \bar{A}$, where

$$p_H \left(R - \frac{B}{\Delta p} \right) = I - \bar{A},$$

then $C = 0$.

By contrast, firms with weaker balance sheets, i.e., $A < \bar{A}$, need to pledge collateral in order to raise funds.[22] Under collateral pledging, the incentive constraint becomes[23]

$$(\Delta p)(R_b + C) \geqslant B,$$

since the borrower loses both her reward R_b and the collateral when she fails (her stake is just larger). The investors' breakeven condition becomes

$$p_H(R - R_b) + (1 - p_H)\beta C \geqslant I - A,$$

or, using the incentive compatibility constraint,

$$p_H \left(R - \frac{B}{\Delta p} \right) + p_H C + (1 - p_H)\beta C \geqslant I - A.$$

Note that the pledging of collateral raises pledgeable income both directly (term $(1 - p_H)\beta C$) and indirectly through the reduction in entrepreneurial reward (term $p_H C$).[24] To minimize the deadweight loss, the borrower pledges the minimum amount of collateral that allows investors to break even:

$$C(A) = \frac{(I - A) - p_H(R - B/\Delta p)}{p_H + (1 - p_H)\beta}.$$

Note that $C(A)$ is a decreasing function of A: among firms that pledge collateral, those with the weakest balance sheet pledge more collateral. […]

22. We assume that $C^{\max}$ is small enough that the NPV remains positive even if the borrower pledges all assets: $p_H R - I - (1 - p_H)(1 - \beta)C^{\max} \geqslant 0$.

23. A different way of writing this constraint is $p_H R_b + (1 - p_H)(-C) \geqslant p_L R_b + (1 - p_L)(-C) + B$.

24. This latter term (and the validity of the analysis) rests on the condition that $R_b \geqslant 0$, which we will assume (this is guaranteed by imposing $B/\Delta p \geqslant C^{\max}$). For large amounts of collateral, it is no longer possible to substitute collateral for reward, since the latter would become negative and violate limited liability.

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14. Dealing with asset substitution

Source: Tirole, Ch. 7, Exercise 7.3, pp. 323–324 (PDF 336) · Answer in book: yes, p. 602

Exercise 7.3 (dealing with asset substitution). Consider the fixed-investment model with a probability that the investment must be resold (redeployed) at an intermediate date because, say, it is learned that there is no demand for the product. The timing is summarized in Figure 7.16.

Figure 7.16 (timeline, left to right; p. 324, PDF 337)

Step 1	Step 2	Step 3 (demand revealed)	Step 4	Step 5
Financing. Firm borrows $I - A$.	Asset substitution?	With probability $1 - x$: verifiable liquidation value L. With probability x: continue $\rightarrow$	Moral hazard (choice of probability of success).	Outcome (R or 0).

An entrepreneur has cash A and wants to invest a fixed amount $I > A$ into a project. The shortfall must be raised in a competitive capital market. The project yields R with probability p and 0 with probability $1 - p$, provided that there is a demand for the product (which has probability x and is revealed at the intermediate stage; the final profit is always 0 if there is no demand, and so it is then optimal to liquidate at the intermediate stage). Investors and entrepreneur are risk neutral, the latter is protected by limited liability, and the market rate of interest is 0.

(i) In a first step, ignore the possibility of asset substitution. The liquidation value is $L = L_0$, and the probability of success is p_H if the entrepreneur works and $p_L = p_H - \Delta p$ if she shirks (in which case she obtains a private benefit B). Assume that the NPV of the project is positive if the entrepreneur works, and negative if she shirks.

Assume that $A \geqslant \bar{A}$, where

$$(1 - x)L_0 + xp_H \left(R - \frac{B}{\Delta p} \right) = I - \bar{A} \qquad (1)$$

(and that $L_0 \leqslant p_H(R - B/\Delta p)$).

- Interpret (1).
- Compute the entrepreneur's expected utility.
- What is the class of optimal contracts (or, at least, characterize the optimal contract for $A = \bar{A}$)?

(ii) Suppose now that, before the state of demand is realized, but after the investment is sunk, the entrepreneur can engage in asset substitution. She can reallocate funds between asset maintenance (value of L) and future profit (as characterized by the probability of success, say).

More precisely, suppose that the entrepreneur chooses L and

- the probability of success is $p_H + \tau(L)$ if the entrepreneur behaves and $p_L + \tau(L)$ if she misbehaves;
- the function τ is decreasing and strictly concave;
- $$\tau(L_0) = 0 \quad \text{and} \quad \tau'(L_0)R = -\frac{1 - x}{x}; \qquad (2)$$
- the entrepreneur secretly chooses L (multitasking).

Consider contracts in which

- liquidation occurs if and only if there is no demand (hence, with probability x);
- the entrepreneur receives $r_b(L)$ if the assets are liquidated, and R_b if they are not and the project is successful (and 0 if the project fails).

Interpret (2). Compute the minimum level of A such that the threat of (excessive) asset substitution is innocuous. Interpret the associated optimal contract. (Hint: what is the optimal asset maintenance (liquidation value)? Note that, in order to induce the entrepreneur to choose this value, in the case of liquidation you may pay $r_b(L) = r_b$ if L is at the optimal level and 0 otherwise.)

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. […] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. […]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. […]

Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. […] There is *credit rationing.* […]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I - A}{p_H} \geqslant R - \frac{I - \bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

[…] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H(R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

So, the borrower's utility jumps up at $A = \bar{A}$. […]

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15. A borrower-friendly bankruptcy court

Source: Tirole, Ch. 4, Exercise 4.9, pp. 190–191 (PDF 203) · Answer in book: yes, p. 586

Exercise 4.9 (borrower-friendly bankruptcy court). Consider the timing described in Figure 4.7.

Figure 4.7 (timeline, left to right; p. 191, PDF 204)

Date 0	Date 1		Date 2
Investment, contract. Entrepreneur has cash A, must borrow $I - A$.	Short-term profit realized. Liquidate → Liquidation value L (known); or Continue →	Moral hazard: Effort decision (high or low).	Final profit: Success (profit R) or failure (profit 0)?

The project, if financed, yields random and verifiable short-term profit $r \in [0, \bar{r}]$ (with a continuous density and *ex ante* mean $E[r]$). After r is realized and cashed in, the firm either liquidates (sells its assets), yielding some known liquidation value $L > 0$, or continues. Note that (the random) r and (the deterministic) L are not subject to moral hazard. If the firm continues, its prospects improve with r (so r is "good news" about the future). Namely, the probability of success is $p_H(r)$ if the entrepreneur works between dates 1 and 2 and $p_L(r)$ if the entrepreneur shirks. Assume that $p'_H > 0$, $p'_L > 0$, and

$$p_H(r) - p_L(r) \equiv \Delta p$$

is independent of r (so shirking reduces the probability of success by a fixed amount independent of prospects). As usual, one will want to induce the entrepreneur to work if continuation obtains. It is convenient to use the notation

$$\rho_1(r) \equiv p_H(r)R \quad \text{and} \quad \rho_0(r) \equiv p_H(r)\left[R - \frac{B}{\Delta p}\right].$$

Investors are competitive and demand an expected rate of return equal to 0. Assume

$$\rho_1(r) > L \quad \text{for all } r \qquad (1)$$

and

$$E[r] + L > I - A > E[r + \rho_0(r)]. \qquad (2)$$

(i) Argue informally that, in the optimal contract for the borrower, the short-term profit and the liquidation value (if the firm is liquidated) ought to be given to investors.

Argue that, in the case of continuation, $R_b = B/\Delta p$. (If you are unable to show why, take this fact for granted in the rest of the question.)

Interpret conditions (1) and (2).

(ii) Write the borrower's optimization program.

Assume (without loss of generality) that the firm continues if and only if $r \geqslant r^*$ for some $r^* \in (0, r)$. Exhibit the equation defining r^*.

(iii) Argue that this optimal contract can be implemented using, inter alia, a short-term debt contract at level $d = r^*$. Interpret "liquidation" as a "bankruptcy."

How does short-term debt vary with the borrower's initial equity? Explain.

(iv) Suppose that, when the decision to liquidate is taken, the firm must go to a bankruptcy court. The judge mechanically splits the bankruptcy proceeds L equally between investors and the borrower.

Define $\hat{r}$ by

$$\rho_0(\hat{r}) \equiv \tfrac{1}{2}L.$$

Assume first that

$$r^* > \hat{r}$$

(where r^* is the value found in question (ii)).

Show that the borrower-friendly court actually prevents the borrower from having access to financing. (Note: a diagram may help.)

(v) Continuing on question (iv), show that when

$$r^* < \hat{r},$$

the borrower-friendly court either prevents financing or increases the probability of bankruptcy, and in all cases hurts the borrower and not the lenders.

Context needed to solve it The statement defines its own model; "as usual" refers to the fixed-investment model's moral-hazard setup (private benefit B, limited liability, incentive constraint), copied here.

From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. […] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. […]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") […].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

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From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[…] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\mathrm{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$